DECOMPOSER BIOLOGY · CROP RESIDUE AFTER HARVEST

Mycro-Tec® Digester

Selected decomposer organisms for post-harvest residue.



Four selected organisms applied once onto stubble, prunings or crop trash after harvest, with an ammonium sulphate starter, as close to rain as you can. They begin breaking down the structural fibre so the carbon and nutrients in the residue return to the soil ahead of the next crop.

01	02	03
Vegetables	**Orchards**	**Broadacre**
Crop trash left after harvest	Prunings and leaf litter on the orchard floor. Not for nut trees.	Cereal, rice, canola and legume stubble

APPLICATION

CROP GROUP	DIGESTER	STARTER	WATER
01 **Vegetables** Crop trash left after harvest	**1–1.5 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**60–100 L/ha**
02 **Orchards** Prunings and leaf litter on the orchard floor. Not for nut trees.	**1–1.5 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**60–100 L/ha**
03 **Broadacre** Cereal, rice, canola and legume stubble	**1–1.5 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**60–100 L/ha**

Manufacturer's rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. One application after harvest; not for in-crop use.

WHAT IS IN IT

01	02	03	04
Trichoderma harzianum	*Aspergillus niger*	*Streptomyces lydicus*	*Penicillium radicum*
FUNGUS	FUNGUS	ACTINOMYCETE	FUNGUS
$>2\times10^{8}$	$>2\times10^{7}$	$>2\times10^{7}$	$>1\times10^{7}$
CFU/ML	CFU/ML	CFU/ML	CFU/ML
Colonises residue rich in cellulose. Its enzymes begin to break down the fibre.	Produces organic acids and enzymes that work on the tougher, more lignified material.	An actinomycete that breaks down residue alongside the fungi.	Works on the surface of residue and at its damp base, alongside Trichoderma.

Three fungi and one actinomycete. Counts are colony-forming units per millilitre at production, as declared on the label. They act on cellulose, hemicellulose, lignin and pectin.

RESIDUE AND APPLICATION CONDITIONS

When Digester is relevant.

Talk to your Hybrid-Ag agronomist about the residue left after harvest and the nutrition required for the next crop. They can advise whether Digester and a prescription blend are appropriate.

01

Residue that will still be there at sowing

Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.

02

Heavy or lignified loads

Big stubble loads, or material that has sat through a dry summer and stalled.

03

Applying close to rain

Applied soon after harvest and as close to rain as you can, before or after it. The breakdown runs on moisture in the residue.

04

Soil and water management

Soil and water conditions affect residue breakdown. Discuss these alongside your nutrition requirements.

What slows it down. Hot, dry spells. Activity falls away through prolonged heat and dryness. Keep out of the heat of the day and line the application up with rain. **Time in the tank.** Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles. **Nitrogen on its own.** Nitrogen feeds the biology already there; it does not add decomposers. The 8 kg of ammonium sulphate is the starter. Do not add more nitrogen to the mix.

Checking residue breakdown. Check for softer stalks at the damp base and residue that pulls away from the soil more easily. Discuss what you find with your agronomist.

SAY

- Supports biological breakdown of crop residue.
- Acts on the structural parts of residue: cellulose, hemicellulose, lignin and pectin.
- One application after harvest, as close to rain as you can.

NOT A CLAIM WE MAKE

- Removes all residue, or guarantees it will disappear.
- Controls stubble disease. Competition for residue is not a disease position.
- Works in-crop, on nut trees, or without the starter.

MIXING ORDER

- Shake the drum well. Add Digester to cool, clean, non-chlorinated water first (neutral pH, not below 5.5), then add the ammonium sulphate and stir thoroughly.
- Strain as you pour. Keep nozzles and filters at 150 µm or larger.
- Mix and apply the same day. Left in the tank overnight it grows mycelia and blocks nozzles.

STORAGE

- Store at 5–40 °C in a clean, dry place out of direct sunlight, drums upright and closed. Do not freeze.
- Twelve months from the date of manufacture. Once a drum is open, use it rather than keep it: the organisms start to germinate and expire without nutrient. Never store the diluted mix.
- Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

SAFETY

- Keep out of reach of children; may irritate skin or eyes, so wash after handling.
- Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.
- Not certified organic. Allowable-input status is assessed case by case with your certifier.

Talk to us about Digester.

Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.

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